



## Team Details

Team Name:

BYTE BRIGADE

| SR. NO | ROLE        | NAME          | ACADEMIC YEAR |
|--------|-------------|---------------|---------------|
| 1      | Team Leader | Prakhar Bajaj | 2nd Year      |
| 2      | Member 1    | Raghav Soni   | 2nd Year      |
| 3      | Member 2    | Parth Kumar   | 2nd Year      |
| 4      | Member 3    | Mayank Lodhi  | 2nd Year      |

**i** A team can have up to 4 members including the team leader. Add rows if necessary.

 COLLEGE NAME

LNCT University, Bhopal

 TEAM LEADER CONTACT NUMBER

+91 91796 86667

 TEAM LEADER EMAIL ADDRESS

prakharbajaj1551@gmail.com



## Problem Statement Addressed



### PS01 - KLA: AI-Based Restoration of Degraded Images for Semiconductor Inspection

#### DESCRIPTION / DETAILS

*Wafer inspection images arrive noisy and low-resolution: fast, low-dose scanning trades image quality for throughput.*

*Three degradations are present - speckle noise (scales with brightness), additive Gaussian sensor noise, and 2x downsampling - applied in an undisclosed order.*

*Real defects hide inside this noise. A missed defect can cost an entire wafer; a false alarm wastes engineer time.*

*Task: learn a mapping from a 128x128 degraded image to its clean 256x256 original, scored on PSNR, SSIM, LPIPS and end-to-end speed.*

CLEAN 256x256

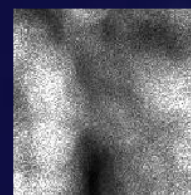


ground truth

speckle noise  
+ Gaussian noise  
+ 2x downsample

→  
applied in an  
undisclosed order

DEGRADED 128x128



what the model receives



## Idea Description - Describe your Idea/Solution/Prototype



Provide a brief summary of your idea, including the key concept and approach. Provide a brief overview of your solution and how it addresses the problem statement.

### KEY CONCEPT & APPROACH

*One small network denoises and 2x super-resolves in a single pass.  
Supervised on the 3,200 provided pairs - no external data, no pretrained restoration weights.  
Predicts only a correction on a bicubic upscale: recovers structure, cannot invent it.*

### SOLUTION OVERVIEW

*NAFNet-SR: 0.53 M parameters, 6.5 MB - runs on a laptop CPU.  
Trained on the judged metrics: Charbonnier + SSIM + LPIPS loss.  
28.00 dB PSNR / 0.759 SSIM / 0.175 LPIPS - beats bicubic on every metric.  
End-to-end 25 ms per image on a T4 GPU (40 images/second).*



## Proposed Solution - Describe your Idea/Solution/Prototype

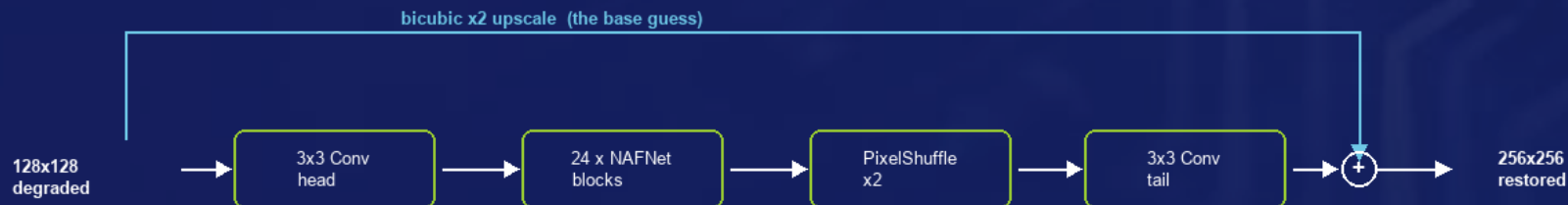


Describe your idea in detail. Include the methodology, technologies involved, and how it addresses the chosen problem statement.

### SOLUTION DETAILS

*WHY NAFNet - SOTA quality, no transformer, no activation function. LOSS - Charbonnier + 0.15 SSIM + 0.05 LPIPS (no GAN loss).*

*TRAINING - 3,000/100/100 seeded split, 128 px crops, batch 16, 100 epochs, ~2 h on a free Colab T4. INFERENCE - directories in/out, batched, tiled.*



**the network learns only the RESIDUAL CORRECTION**  
noise to remove and edges to sharpen - not the whole image



# Innovation and Uniqueness



Highlight what makes your idea unique or innovative compared to existing solutions.

## KEY INNOVATION

*We train directly on the scoreboard: SSIM and LPIPS terms sit inside the loss, so the model optimises two of the three judged metrics rather than hoping they follow from pixel accuracy.*

*Hallucination is strongly constrained: the global bicubic skip means the network predicts a correction rather than generating an image freely.*

*Float-faithful pipeline: degraded inputs legitimately exceed  $[0,1]$  and we never clip them; clipping is applied only to outputs, where ground truth guarantees the range.*

## COMPETITIVE ADVANTAGE

*SMALLER: 0.53 M parameters versus 10-26 M for Restormer or SwinIR, so throughput - an explicit scoring axis - is a strength, not a compromise.*

*FASTER: 25 ms per image end-to-end on a T4; all 400 test images in ~10 s.*

*TRUSTWORTHY: beats the bicubic baseline on 100 of 100 held-out images; failure cases are published and analysed rather than hidden.*

*REPRODUCIBLE: seeded splits, one-command training and inference, weights and runtime report committed to the repository.*





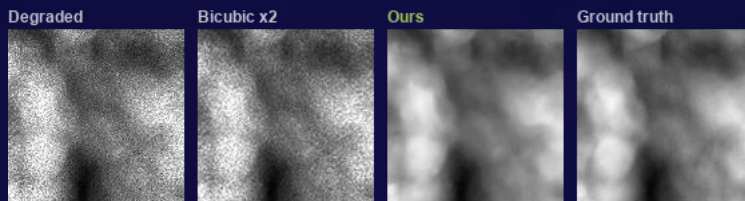
# Impact and Benefits



Explain how your solution will make an impact, such as improving performance, reducing costs, increasing efficiency, or solving other challenges.

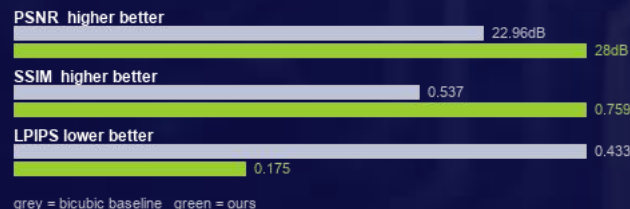
## Primary Impact

Detail hidden inside sensor noise becomes visible again - fewer missed defects and fewer false alarms.  
Enables faster or lower-dose scanning: recover quality in software, so existing microscopes inspect more wafers.



## Quantifiable Outcomes

PSNR 22.96 -> 28.00 dB (+5.04 dB)  
SSIM 0.537 -> 0.759 | LPIPS 0.433 -> 0.175  
25 ms/image - all 400 test images in ~10 s  
0.53 M parameters, 6.5 MB - runs on CPU





## Technology & Feasibility/Methodology Used



Describe the technologies, methodologies, or tools you plan to use to implement your idea.

### IMPLEMENTATION STRATEGY

*Trained end to end on a free Google Colab T4 in about two hours; no paid infrastructure was used at any stage.  
Inference runs on GPU or CPU. Batched on CUDA, tiled for large images, and verified on 128, 256 and non-square inputs.  
Deployment is a drop-in folder-to-folder script plus a Streamlit UI for operators; evaluators need no source-code edits.  
Measured, not estimated: benchmark.py reports end-to-end runtime with hardware and software versions stated.*



PyTorch 2.11 - NAFNet-SR



NVIDIA T4 / H100 - CPU  
capable



Colab - Git - Streamlit



## GitHub & Video Link



### GitHub Repository

[github.com/prakharbajaj1551/semicon-image-restoration](https://github.com/prakharbajaj1551/semicon-image-restoration)



### Prototype / Simulation Video

<https://www.youtube.com/watch?v=RMSDaviTOlw>





# Research and References



## Research Background & Methodology

Briefly describe the research foundation or scientific principles supporting your idea.

*Built on published, peer-reviewed research rather than an unproven design: NAFNet (ECCV 2022) matches transformer restorers at a fraction of the cost.*

*Loss follows restoration literature - Charbonnier for pixel fidelity, SSIM for structure, LPIPS for perceptual quality.*

*Validated on a held-out split with no selection leakage, against a bicubic baseline, with failure cases analysed.*



## References & Citations

List key papers, articles, or data sources.

Ref 1: Chen et al., Simple Baselines for Image Restoration (NAFNet), ECCV 2022 - [arxiv.org/abs/2204.04676](https://arxiv.org/abs/2204.04676)

Ref 2: Zhang et al., The Unreasonable Effectiveness of Deep Features as a Perceptual Metric (LPIPS), CVPR 2018

Ref 3: Shi et al., Real-Time Single Image Super-Resolution Using an Efficient Sub-Pixel CNN, CVPR 2016